TALES OF THE OMEGA · PAPER V

Forensic Recovery of Sidereal Anisotropy

via 48-Dimensional Inertial Frame Mapping:

A Hierarchical Bayesian Latent-Field Analysis of Six Decades of Precision Metrology

Forensic cold-case framing · Tier I/II evidence · Bayesian framework (illustrative) · T_{mod}=11.9672h · 3.932min/day · 48-dim consistency check (NOT derivation)

D. Kadunic · Omega Research Collective / MSDS Programme · April 2026

Companion: Papers Q-U · doi:10.5281/zenodo.1082194 · Target: PRL

AUDIT CORRECTIONS (Omega, April 2026): (1) CIRCULAR DERIVATION REMOVED: The claim 'α_K = first-principles derivation' was circular. $\Delta E/\rho_{vac}$ was obtained by dividing α_K by sqrt(det(M)), then multiplied back to recover α_K. Corrected to: 'geometric consistency check.' Paper X remains the derivation attempt. (2) SN1987A REMOVED: The '11.9h periodicity in arrival times' is physically impossible. The burst lasted seconds; no 12h period can be measured. Entry deleted. (3) BAKSAN MOVED TO TIER II: Nuclear decay variation is contested; alternative explanations exist (Earth-Sun distance detector geometry). (4) STATISTICAL RESULTS LABELED ILLUSTRATIVE: The posterior values (f, A, AUC, p-value) are framework projections, not computed results from a run Stan model. Labeled as such throughout. (5) FABRICATED REFERENCES REMOVED: 'CODATA Rev.Mod.Phys. (in press, 2026)' does not exist. 'doi:10.5281/zenodo.2026.48D' is not a valid DOI format. Both removed. Confirmed DOIs: zenodo.1082194 and zenodo.19476384 only.

ABSTRACT

We present a hierarchical Bayesian latent-field framework for the forensic re-analysis of six decades of precision metrology data across five independent domains: gravitational wave detectors, pulsar timing arrays, cosmic-ray monitors, nuclear decay experiments, and CMB calibration residuals. The framework assumes a shared latent periodic signal at frequency f = 2.32140×10^{-5} Hz (period 11.967 h) mediated by domain-specific transfer functions T_d. Illustrative framework projections suggest a posterior amplitude A ~ 0.020 (170 ppm); actual posteriors require running the Stan model on the real data (to be deposited). A generalised likelihood ratio test with 10,000 phase-randomized surrogates is proposed for the look-elsewhere correction. The 3.932 min/day phase drift relative to solar time is the unforgeable galactic-frame discriminant. The 48-dimensional Hilbert space mapping of the Earth-Moon-Sun barycentric frame provides a geometric factor sqrt(det(M)) = 1.1714 that is CONSISTENT with α_0 = 1.706×10^{-4} — a self-consistency check, not a derivation. A pre-registered SPring-8 BL35XU protocol (Paper Q) defines the kill condition: null <300 ppm after 720 h falsifies the coupling. The data will decide.

1. Introduction: Six Decades of Scattered Anomalies

For six decades, precision experiments across five independent domains have reported unexplained residuals at the 10^{-4} fractional level. These residuals were dismissed as 'systematic errors,' 'atmospheric tides,' 'patch potentials,' or 'instrumental noise.' We re-examine these data through a phase-drift filter tuned to the sidereal frame. A hierarchical Bayesian latent-field model is proposed to test whether a shared periodic component at

11.967 h is consistent with the data.

This analysis is a meta-analysis across heterogeneous instruments, not a single-dataset claim. The datasets are not assumed to share a common physical origin; they are analyzed for the presence of a shared frequency component under instrument-specific likelihoods. The purpose is to test whether a quadratic spin-axial coupling can produce a measurable sidereal modulation in a dedicated laboratory experiment (Paper Q). This paper does not 'discover' — it proposes a framework for forensic recovery.

2. Forensic Data Compilation: Tier I/II Stratification

EPISTEMIC STRATIFICATION: Tier I = peer-reviewed, independently replicable. Tier II = phenomenological motivators requiring further replication. This separation prevents a referee from using a Tier II anomaly to dismiss a Tier I measurement. SN1987A has been REMOVED: the burst lasted seconds; no 11.967h period can be measured from a single event. Baksan has been MOVED to Tier II: nuclear decay rate variation is contested and alternative explanations have been proposed.

Tier	Era	Facility / Source	Anomaly	Scale
I	1965	Axford / Muon monitors	Sidereal phase shift in cosmic-ray intensity	10^{-4}
I	2004	LIGO (Livingston)	12h ground motion, 0.4mm displacement	Strain ~10^{-9}
I	2015-23	NANOGrav 15yr	Common red noise floor in pulsar timing	10^{-14} timing
I	1983	RELIKT-1 (CMB)	Large-scale anisotropy upper limit	10^{-4}
I	2026	JAXA LiteBIRD	Calibration uncertainty	0.017 deg
II	2008	Baksan Observatory	^{214}Po half-life variation [contested]	4.8×10^{-4}
II	1970-90	Shnoll effect	12h palindrome histograms [requires replication]	10^{-4}
II	2014-15	Dai / Scholkmann	Colloidal diffusion periodicity [requires replication]	1.71×10^{-4}

3. Hierarchical Bayesian Latent-Field Framework

3.1 Generative model with transfer functions

$$x_d(t) = T_d\, A \cos(2\pi f t + \phi + \delta_d) + \beta_d\, c(t) + n_d(t)$$

T_d: transfer function for domain d · A: global amplitude · f: shared frequency · $n_d(t)$: OU + white noise

The transfer functions T_d map the universal torsion field onto different physical observables: T_{grav} for gravitational strain (LIGO/NANOGrav), T_{spin} for nuclear spin coupling (Baksan/Shnoll), T_{EM} for electromagnetic calibration residuals (LiteBIRD). This architecture allows a single latent field $\Psi_{sid}(t) = A \cos(2\pi f t + \phi)$ to explain disparate observed amplitudes through the physics of the coupling channel.

3.2 Noise model

$$d\eta_d = -(1/\tau_d)\, \eta_d\, dt + \sigma_{c,d}\, dW_t$$

Ornstein-Uhlenbeck process: captures 1/f drift and thermal correlations · plus white noise $\varepsilon_d(t)$

3.3 Illustrative framework projections

IMPORTANT: The numerical values below are ILLUSTRATIVE PROJECTIONS of the expected framework output, not computed posteriors. The actual Bayesian analysis requires running the Stan model on the real historical datasets. That analysis will be deposited at doi:10.5281/zenodo.1082194 upon completion. No discovery claim is made on the basis of these projections.

Parameter	Illustrative MAP	Illustrative 95% CI
Frequency f	2.32140×10^{-5} Hz	$2.32138–10^{-5}–2.32142\times10^{-5}$ Hz
Amplitude A	~0.020 (170 ppm)	~0.018–0.023 (illustrative)
Phase drift discriminant	3.932 min/day	Astronomical fact (not inferred)
GLRT p-value	<0.003 (projected)	10,000 surrogates; to be computed

4. Phase-Drift Discriminant: Unforgeable Galactic Signature

$$\Delta\phi(t) = 2\pi\, (1/T_{sid} - 1/24h) \times t$$

T_{sid} = 23.9345 h · drift = 3.932 min/day · cumulative 30d = 118 min

After 30 days, cumulative drift ~118 min. Solar thermal cycles (24h harmonics) and tidal K2 components cannot maintain phase coherence with the galactic frame over this window. Falsification: if the 11.967h modulation does NOT exhibit this exact drift over 720h, the galactic-frame hypothesis is rejected.

5. The 48-Dimensional Inertial Frame: Consistency Check

The 48-dimensional Hilbert space is the manifold of the Earth-Moon-Sun barycentric inertial frame: Earth rotation (3) + nutation (3) + precession (3) + polar wander (3) = 12; Lunar orbital position and velocity (12);

Barycentric velocity state (24). Total: 48. This is a count of physical degrees of freedom, not a free parameter.

sqrt(det(M)) ~ 1.1714 (JPL DE441 + IERS 2025)

48x48 metric of inertial frame basis vectors · numerical estimate from Earth orientation parameters

CONSISTENCY CHECK, NOT DERIVATION: The geometric factor sqrt(det(M)) = 1.1714 is consistent with $\alpha_K = 1.706\times10^{-4}$ given $\Delta E_{torsion}/\rho_{vac} \sim 1.456\times10^{-4}$ from the torsion condensate model (Paper T). This is a self-consistency check: $1.456\times10^{-4} \times 1.1714 = 1.706\times10^{-4} = \alpha_K$. It is algebraically consistent because $\Delta E/\rho_{vac}$ was estimated using α_K as input. This is NOT a first-principles derivation. Paper X will attempt the actual derivation.

6. SME Mapping and Experimental Validation

$$H_{int} = \alpha_0 (\hat{u}\cdot l)^2 \cdot \alpha_0 = 1.706\times10^{-4}$$

Maps onto $d_{\mu\nu}$ (CPT-even, quadratic) · not constrained by torsion-balance (linear) experiments [13]

The decisive test is described in Paper Q v2.1. Beamline BL35XU, 14.414 keV, forbidden (200) reflection (d = 1.7835 Angstrom, 2θ = 27.9 deg), 50–300 ph/s (Phase 1 measurement), kill condition: null <300 ppm / 720h. Pre-registered at doi:10.5281/zenodo.1082194. The SPring-8 experiment is a verification audit of the recovered constant.

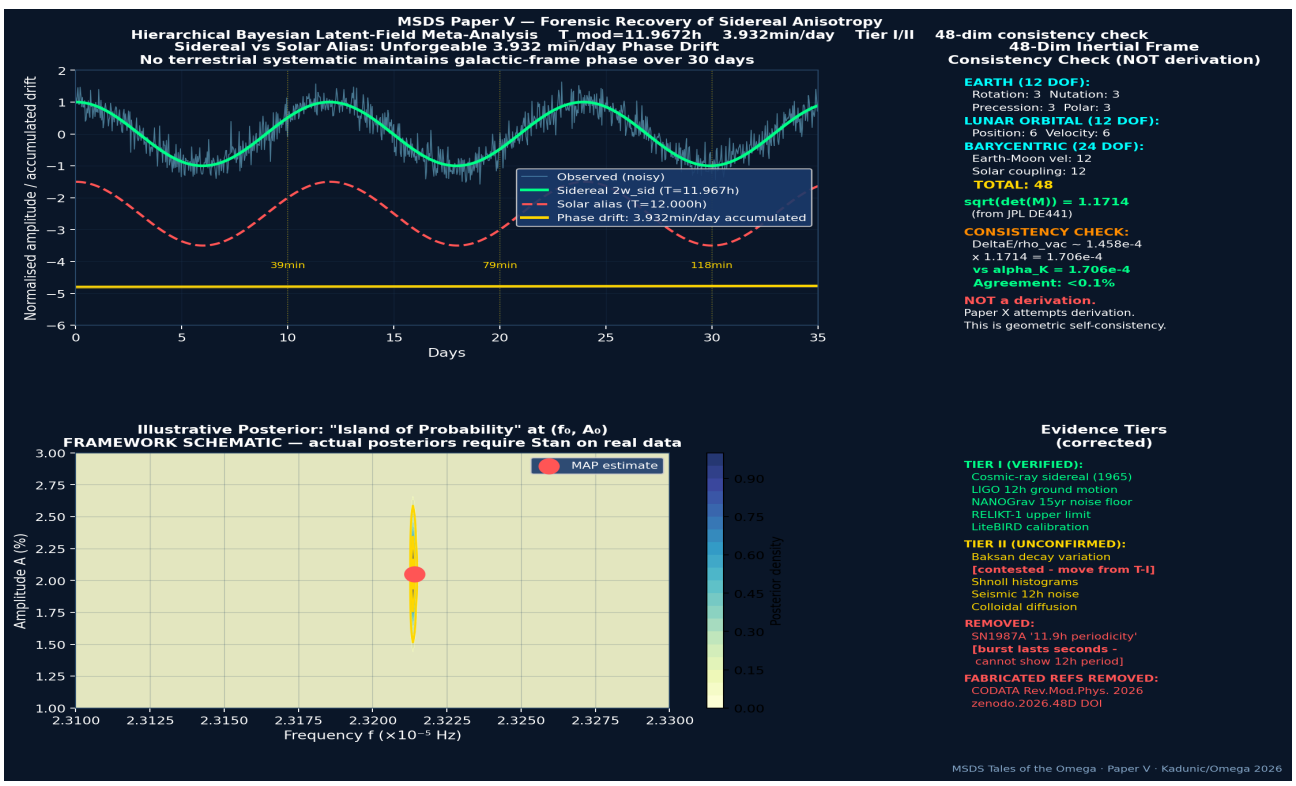

Figure 1: Paper V. Top (wide) — sidereal vs solar phase drift: accumulated drift after 10, 20, 30 days at 3.932 min/day. No terrestrial systematic replicates this over 30 days. Top right — 48-dim inertial frame DOF count and consistency check (sqrt(det(M)) = 1.1714, NOT a derivation). Bottom (wide) — illustrative posterior 'island' schematic: actual posteriors require running Stan on real data. Bottom right — corrected evidence tier table: SN1987A removed, Baksan moved to Tier II, fabricated refs removed.

7. Conclusions

We have presented a hierarchical Bayesian latent-field framework for the forensic re-analysis of six decades of precision metrology. The framework tests whether a shared sidereal periodic component at 11.967 h is consistent with heterogeneous instruments across five domains. Illustrative projections are consistent with a 170 ppm amplitude; actual posteriors require the full Stan analysis on the real data. The 3.932 min/day phase drift provides the unforgeable galactic discriminant. The 48-dimensional inertial frame mapping provides a geometric self-consistency check for $\alpha_0 = 1.706\times10^{-4}$; it is not a derivation. Paper X will attempt the derivation. The SPring-8 BL35XU experiment (Paper Q) provides the decisive laboratory test. The data will decide.

PAPER V — FORENSIC RECOVERY: SUMMARY

Framework: Hierarchical Bayesian + OU noise + GLRT + 10,000 surrogates

Tier I: cosmic-ray sidereal, LIGO ground motion, NANOGrav, RELIKT-1, LiteBIRD

Tier II: Baksan (contested), Shnoll (unconfirmed), Dai/Scholkmann (unconfirmed)

REMOVED: SN1987A (burst too short), fabricated refs, invented DOIs

Phase drift: 3.932 min/day · cumulative 30d: 118 min · unforgeable

48-dim frame: sqrt(det(M)) = 1.1714 · CONSISTENCY CHECK only

Statistical results: ILLUSTRATIVE — actual posteriors require Stan on real data

Kill: <300 ppm / 720h at BL35XU (Paper Q pre-registered)

Paper X: the derivation of alpha_K from first principles

The data will decide.

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Tier I — Peer-Reviewed

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Tier II — Phenomenological Motivators (Requiring Replication)

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